

WHAT MATTERS FOR REPRESENTATION ALIGNMENT: GLOBAL INFORMATION OR SPATIAL STRUCTURE?

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ABSTRACT

Representation alignment (REPA) improves generative training by transferring representations from a pretrained vision encoder into intermediate diffusion features. We revisit a broad design question that arises whenever external visual representations are used for generation: which property of the target representation is most useful, its *global* semantic information (e.g., measured by ImageNet-1K accuracy) or its *spatial* structure (i.e. pairwise cosine similarity between patch tokens)? A common interpretation of prior results is that stronger global semantic performance should make an encoder a better alignment target. We examine this assumption through a large-scale empirical study spanning 27 vision encoders and multiple generative model scales. Across these settings, the dominant predictor of generation performance is spatial structure rather than global performance. Motivated by this observation, we introduce two minimal modifications that more directly preserve and transfer *spatial* information during alignment. Specifically, we replace the standard MLP projection layer in REPA with a simple convolution layer and apply a spatial normalization layer to the external representation. The resulting method, iREPA, requires fewer than four lines of code and consistently improves REPA convergence across a diverse set of vision encoders, model sizes, and training variants (including REPA, REPA-E, Meanflow, and JiT). These results suggest that spatially structured target representations are broadly useful alignment signals for generative models, and motivate revisiting how representation alignment objectives are designed and evaluated.

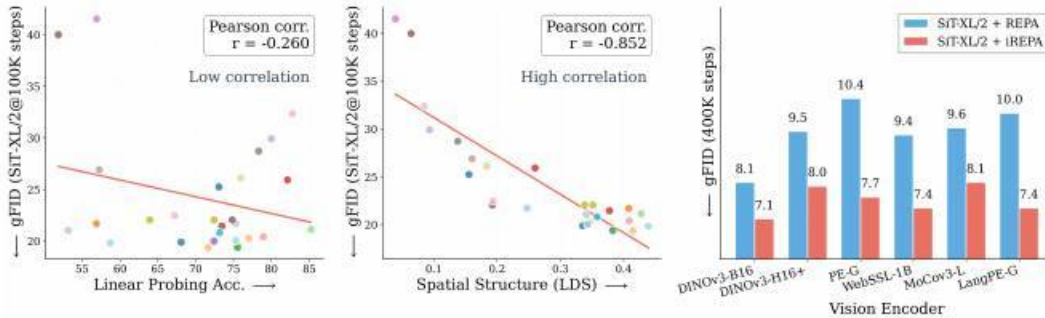


Figure 1: **What matters for representation alignment?** *Left:* Correlation analysis across 27 diverse vision encoders. Across this encoder sweep, spatial structure is a stronger predictor of target-representation performance for generation than global performance measured by linear probing accuracy. *Right:* We further test this insight with two simple modifications that accentuate transfer of spatial features from the target representation to the diffusion model. The resulting approach improves REPA convergence across the evaluated encoders, model sizes, and training variants.

1 INTRODUCTION

Representation alignment has become a general strategy for accelerating diffusion-transformer training across modern image-generation pipelines (Ma et al., 2024; Peebles & Xie, 2023). Rather than training diffusion features in isolation, these methods align intermediate representations with



Figure 2: **Motivating examples — spatial structure matters.** Metrics comparison showing inverse relationship between ImageNet accuracy and generation quality. *Left:* PE-G, despite having significantly higher validation accuracy (82.8% vs. 53.1%), shows worse generation quality compared to SpatialPE-B (Bolya et al., 2025). *Right:* Similarly, WebSSL-1B (Fan et al., 2025) also shows much better global performance (76.0% vs. 53.1%), but worse generation. **Spatial Self-Similarity:** We find that spatial structure instead provides a better predictor of generation quality than global performance. See §3 for spatial structure metric. All results reported at 100K using SiT-XL/2 and REPA.

pretrained self-supervised visual encoders, yielding faster convergence and stronger final performance in recent systems (Yu et al., 2024; Leng et al., 2025a). Yet the mechanism by which an external encoder helps a generative model remains under-specified. In particular, it is unclear whether the useful signal is primarily the encoder’s global semantic content, often summarized by linear probing performance, or the spatial organization of patch tokens, captured by relationships among local representations.

Clarifying this distinction matters beyond any single encoder choice, because representation alignment is increasingly used as a reusable training component for generative models. A common working assumption is that a better alignment target is one with stronger ImageNet-1K validation accuracy, treating that accuracy as a proxy for global semantic understanding (Oquab et al., 2024; Chen et al., 2021). Under this view, target representations with better ImageNet performance are expected to induce better generation (Yu et al., 2024). Consequently, higher linear probing accuracy of diffusion features is often interpreted as evidence that alignment is working, reinforcing a broad emphasis on global semantic information as the central explanatory factor. The following quote from REPA (Yu et al., 2024) summarizes this interpretation:

“When a diffusion transformer is aligned with a pretrained encoder that offers more semantically meaningful representations (i.e., better linear probing results), the model not only captures better semantics but also exhibits enhanced generation performance, as reflected by improved validation accuracy with linear probing and lower FID scores.”

We revisit this assumption by asking a more general question about external representations for generation: when an encoder is used as an alignment target, which aspect of its representation transfers most effectively? Through a large-scale empirical analysis over diverse vision encoders, including recent large vision foundation models such as WebSSL (Fan et al., 2025), DINOv3 (Siméoni et al., 2025), perceptual encoders (Bolya et al., 2025), and C-RADIO (Heinrich et al., 2024), we identify three findings that consistently broaden the role of spatial structure in representation alignment.

Higher validation accuracy does not imply better representation for generation. Across the studied encoders, stronger global semantic performance, measured by ImageNet-1K linear probing accuracy, does not reliably translate into a better generative alignment target. For instance, PE-Spatial-B, a spatially tuned model derived from PE-Core-G (Bolya et al., 2025), has much lower validation accuracy¹ on patch tokens than PE-Core-G (53.1% vs. 82.8%) but produces better generation with REPA (Figure 2). Likewise, WebSSL-1B (Fan et al., 2025) has much stronger global performance (76.0% vs. 53.1%) than PE-Spatial-B but yields worse generation in this alignment setting. In Section §2, we show that this mismatch appears across many target representations, suggesting that the benefits of representation alignment are not governed solely by global classification-oriented semantics.

¹Similar to REPA (Yu et al., 2024), we use linear probing accuracy on patch tokens to measure global semantic performance of external representation as only patch tokens are used for representation alignment.

Spatial structure, not global information, determines generation performance. To make this trend measurable, we evaluate several simple metrics for the spatial self-similarity structure (Shechtman & Irani, 2007) among patch tokens (§3). We then correlate generation FID under REPA with both linear probing accuracy and spatial structure over 27 vision encoders and 3 model sizes (SiT-B, SiT-L, SiT-XL). Across this benchmark sweep, all spatial structure metrics are substantially more correlated with generation FID (Pearson $|r| > 0.852$) than ImageNet-1K validation accuracy ($|r| = 0.26$). Additional experiments support the same broader conclusion: representations with limited global information can still provide useful alignment signals when they preserve spatial organization. For example, SAM2 improves generation with REPA more than encoders with roughly $\sim 60\%$ higher ImageNet accuracy (Fig. 3), and classical spatial descriptors such as SIFT (Lowe, 1999) and HOG (Dalal & Triggs, 2005), while not the strongest targets, still provide meaningful gains under representation alignment (§3).

Accentuating spatial features improves convergence performance. Building on these findings, we introduce two lightweight changes (< 4 lines of code) that make REPA transfer spatial information from the target representation more directly. First, we add a spatial regularization layer that increases spatial contrast in the target representation. Second, because the standard MLP projection used to map diffusion features to target dimensions can weaken spatial information (Figure 6a), we replace it with a simple convolution layer. The resulting method, iREPA, consistently accelerates REPA convergence across the studied range of encoders, diffusion model sizes, and training recipes, including REPA-E (Leng et al., 2025a) and Meanflow (Geng et al., 2025) w/ REPA.

We summarize the main contributions of this paper below:

- We provide a large-scale empirical analysis showing that spatial structure, rather than global semantic accuracy alone, better explains the effectiveness of representation alignment.
- We introduce a straightforward and fast-to-compute Spatial Structure Metric (SSM), which correlates more strongly with downstream FID performance than linear probing scores across the studied encoders and model scales.
- We propose simple modifications that emphasize spatial transfer from the target representation to diffusion features. The resulting method improves REPA convergence across target representations, model architectures, and training recipe variants (REPA, REPA-E, Meanflow w/ REPA, JiT w/ REPA etc).

2 MOTIVATION: GLOBAL INFORMATION MATTERS LESS

We begin with motivating cases that test a broad assumption in representation alignment: a target representation with higher global performance, measured by ImageNet-1K accuracy, should also be a better target for generation with REPA. Across several encoder families and controlled manipulations, this assumption does not hold consistently. We then show in §3 that the same observations are better accounted for by measuring the spatial structure of the target representation.

Recent vision encoders. We compare recent vision encoders spanning Perceptual Encoders (Bolya et al., 2025), WebSSL (Fan et al., 2025), and DINOv3 (Siméoni et al., 2025). PE-Spatial-B (80M), a small spatially tuned model derived from PE-Core-G (1.88B) (Bolya et al., 2025), illustrates the broader mismatch between global accuracy and alignment utility. As shown in Figure 2, PE-Core-G has much higher ImageNet-1K accuracy (82.8% vs 53.1%) but performs worse as the REPA target representation (FID 32.3 vs 21.0). WebSSL-1B (1.2B) follows a similar pattern: despite much higher ImageNet-1K accuracy (76.0% vs 53.1%), it is a weaker REPA target than PE-Spatial-B (FID 26.1 vs 21.0).

SAM2 outperforms vision encoders with much higher ImageNet-1K accuracy. To examine a target with limited global classification information, we evaluate the SAM2-S vision encoder (46M) (Ravi et al., 2024), whose validation accuracy is only 24.1% (Fig. 3a). When used with REPA, SAM2-S nevertheless achieves better FID than several encoders with substantially higher ImageNet-1K accuracy, including PE-Core-G (82.8%). This case suggests that useful alignment signals can come from properties not captured by global classification accuracy.

Larger models within same encoder family can have similar or worse generation performance. Within a fixed encoder family, larger variants are often expected to provide better representations because they tend to improve ImageNet-1K accuracy. For representation alignment, however, larger

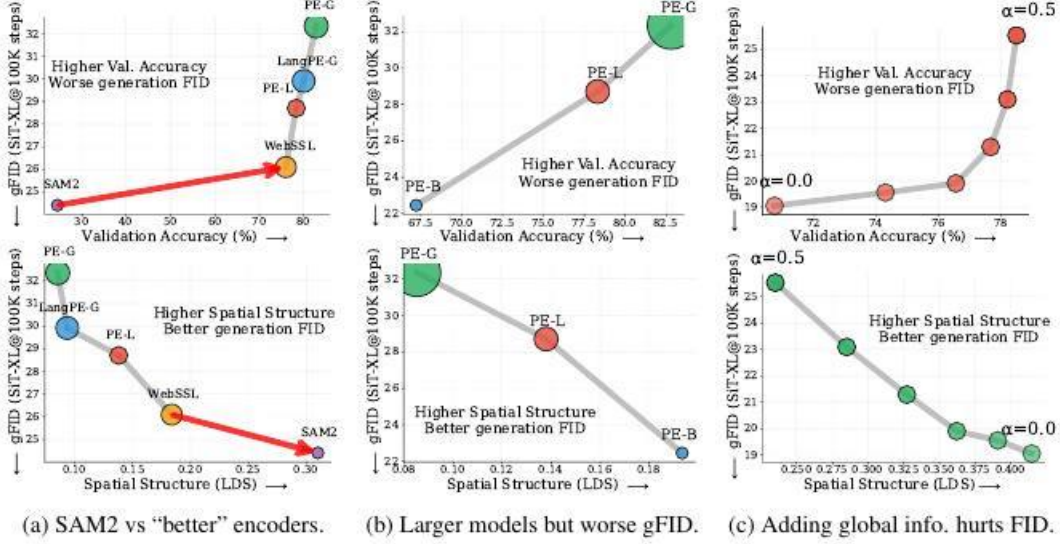


Figure 3: **Higher global information does not imply better REPA performance.** **Top row:** Several trends showing global performance does not correlate well with generation FID when using REPA. (a) SAM2-S with only validation accuracy of 24.7% results in better generation performance with REPA compared with other models with $\sim 60\%$ higher validation accuracy. (b) Larger encoders within same family can have better validation but worse generation performance. (c) Adding global information to patch tokens via CLS token improves global performance but hurts generation. **Bottom row:** We show that spatial structure rather than global performance provides a better indicator for generation. Please see §3 for large-scale detailed analysis across different spatial structure metrics. All results are reported w/o classifier free guidance, SiT-XL/2 w/ REPA and 250 NFE (Yu et al., 2024) for inference.

variants can produce similar (DINOv2) or even worse (PE, Cradio) generation performance (Fig. 3b). Notably, (Yu et al., 2024) report a related trend for DINOv2, explaining it as “we hypothesize is due to all DINOv2 models being distilled from the DINOv2-g model and thus sharing similar representations”. In §3, we show that spatial structure provides a more general explanation for this family-level behavior.

Adding more global information can hurt generation. We further isolate the role of global information through a controlled DINOv2 experiment that injects the CLS token into local patch tokens. The CLS token mixing operation is $\mathbf{p}_i^{\text{new}} = \mathbf{p}_i + \alpha \cdot \mathbf{c}$, where $\mathbf{p}_i$ denotes the i^{th} patch token, $\mathbf{c}$ the CLS token, and $\alpha \in [0, 0.5]$ controls the mixing strength. As shown in Figure 3c, increasing α from 0 to 0.5 monotonically improves linear probing accuracy from 70.7% to 78.5%. At the same time, generation quality deteriorates, with FID worsening from 19.2 at $\alpha = 0$ to 25.4 at $\alpha = 0.5$.

Together, these examples show that global performance is not a reliable standalone indicator of REPA performance across encoder choices, encoder scales, or controlled feature manipulations. We next show in §3 that spatial structure provides a stronger and more broadly applicable signal.

- **Finding 1.** Better global semantic information (e.g., ImageNet-1K accuracy) does not imply a better representation for generation.

3 SPATIAL STRUCTURE MATTERS MORE

We hypothesize that spatial structure, rather than global information alone, is the representation property that most broadly predicts generation performance under REPA. To quantify this property, we use simple metrics for the spatial self-similarity structure (Shechtman & Irani, 2007) of target representations. We then show that these spatial metrics correlate more strongly with gFID than ImageNet-1K accuracy and also explain the encoder-level and intervention-level trends introduced in §2.

Measuring spatial self-similarity structure. Given an image $\mathcal{I}$ and a vision encoder $\mathcal{E}$, we define the quantities needed to measure spatial organization across patch tokens:

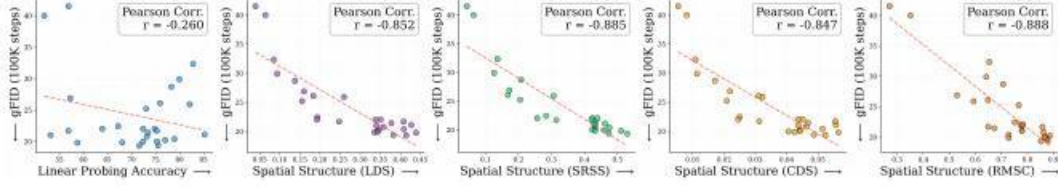


Figure 4: Spatial structure correlates more strongly with generation quality than linear probing. Correlation analysis across 27 diverse vision encoders using SiT-XL/2 and REPA. Linear probing is weakly correlated with FID (Pearson $|r| = 0.260$), whereas spatial structure metrics—LDS ($|r| = 0.852$), SRSS ($|r| = 0.885$), CDS ($|r| = 0.847$), and RMSC ($|r| = 0.888$)—show much stronger correlation with generation performance. See Fig. 10 for detailed plots with encoder labels.

- *Self-similarity.* Let $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ denote the extracted patch representations, with $T = H \times W$ patches. Let the kernel $K_{\mathcal{X}}(\cdot, \cdot) \in \mathbb{R}$ measure appearance self-similarity (Shechtman & Irani, 2007) between patch tokens. Here, we use the cosine kernel $K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2}$.
- *Spatial distance.* Let $\mathcal{P} \in \mathbb{N}^{T \times 2}$ store the spatial location of each of the T tokens in coordinate space, and let $d(\cdot, \cdot) \in \mathbb{N}$ denote the Manhattan distance between token pairs.

We measure *spatial* self-similarity by asking how self-similarity $K_{\mathcal{X}}$ changes with lattice distance d between patch tokens. Larger values indicate stronger spatial organization, meaning nearby patches are more similar to one another than distant patches. Unless otherwise stated, we use a simple correlogram contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}, \mathcal{P}) = \mathbb{E}_{d(t, t') \in (0, r_{\text{near}})} K_{\mathcal{X}}(t, t') - \mathbb{E}_{d(t, t') \geq r_{\text{far}}} K_{\mathcal{X}}(t, t').$$

The final spatial self-similarity score (LDS) is computed as the expectation over patch representation $\mathcal{X}$. We use $r_{\text{near}} = r_{\text{far}} = H/2$ in the main experiments, and the observed correlations are robust to these choices. Appendix B gives the exact implementation details and reports alternative spatial metrics (CDS, SRSS, RMSC), which show the same qualitative behavior (see Fig. 4).

Spatial structure correlates much higher with generation performance than linear probing.

We next perform a large-scale correlation analysis across 27 diverse vision encoders. As shown in Figure 4, the commonly used linear probing score is only weakly correlated with generation performance (Pearson $|r| = 0.26$), whereas all SSM metrics are much more strongly correlated with generation performance (Pearson $|r| > 0.85$).

Generalization across model scales. We further test whether this relationship transfers across generative model scales (SiT-B, SiT-L, SiT-XL) in Figure 5. Linear probing remains weakly correlated across scales ($|r| < 0.306$), while spatial structure maintains a much higher correlation with generation performance ($|r| > 0.826$).

- **Finding 2.** *Spatial structure correlates much higher with generation performance than linear probing. We next use spatial metrics to explain previously unexplained trends.*

Spatial structure can explain previously unexplained trends. As discussed in §2, global performance (ImageNet-1K accuracy) is not a reliable predictor of representation-alignment effectiveness in the studied settings. The spatial metrics above provide a more consistent explanation of those trends.

(1) *PE-Spatial-B vs PE-Core-G:* Figure 2 shows that PE-Core-G achieves much higher ImageNet-1K accuracy (82.8% vs 53.1%) but performs worse as a REPA target representation (FID 32.3 vs 22.0). The spatial metric resolves this apparent contradiction: despite lower global performance, PE-Spatial-B has substantially stronger spatial structure than PE-Core-G, matching its better generation performance in Figure 2.

(2) *SAM2 outperforms “better” vision encoders:* §2 shows that although SAM2 has much lower ImageNet-1K accuracy (24.1%), it yields better generation than encoders with $\sim 60\%$ higher accuracy. Fig. 3a shows that this behavior is consistent with SAM2’s stronger spatial structure.

(3) *Larger models in same encoder family underperform:* As shown in Fig. 3b, larger models in the same encoder family can improve ImageNet-1K accuracy while weakening spatial structure, which

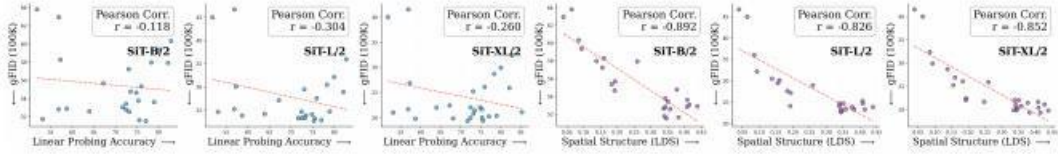
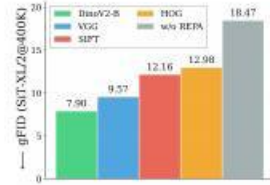


Figure 5: **Correlation analysis across model scales.** Across model scales, spatial structure (right) consistently correlates more strongly with gFID than linear probing (left).

corresponds to worse REPA generation performance. This also aligns with observations from Yu et al. (2024), where larger models can have similar or worse generation performance.

(4) *Adding global information to patch tokens via CLS token hurts generation:* In Figure 3c, increasing global information by mixing the CLS token with patch tokens improves global performance. However, CLS mixing reduces spatial contrast among patch tokens, making otherwise unrelated regions (e.g., foreground object and background) more similar. The resulting loss of spatial structure coincides with worse generation performance.

If spatial structure matters more, can we use SIFT or HOG features for REPA? Surprisingly, yes. Although they are not the strongest targets, classical spatial features such as SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), and intermediate VGG features (Simonyan & Zisserman, 2014) all improve performance with REPA. This supports the broader view that representation alignment can benefit from spatially organized features even when they do not provide rich global semantic information.



Can we use spatial metrics to explain gains with REPA? Yes. The introduced spatial metrics explain both the gains obtained from REPA and the additional improvements from our spatially focused training recipe, iREPA, which we introduce next in §4.



4 iREPA: IMPROVING REPRESENTATION ALIGNMENT BY ACCENTUATING WHAT MATTERS

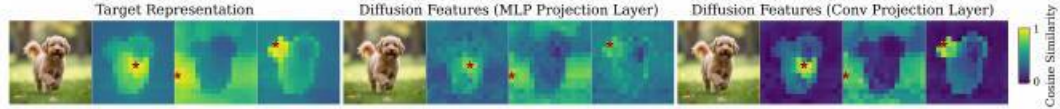
To further test whether spatial structure is a broadly useful alignment signal, we introduce two straightforward modifications to the original REPA training recipe. Both changes are designed to preserve and transfer spatial features from the teacher vision encoder to the student diffusion transformer without changing the underlying datasets, losses, or evaluation protocol.

Convolutional projection layer instead of MLP. The standard REPA approach uses a 3-layer MLP projection to map diffusion feature dimensions to the dimensionality of the external representation. As shown in Figure 6a, this projection can be lossy for spatial organization and can reduce the contrast between patch tokens. We therefore replace the MLP with a lightweight convolutional layer (kernel size 3, padding 1) that operates directly on the spatial grid. This convolutional projection keeps the alignment interface simple while adding a local inductive bias that is better matched to spatially structured target representations.

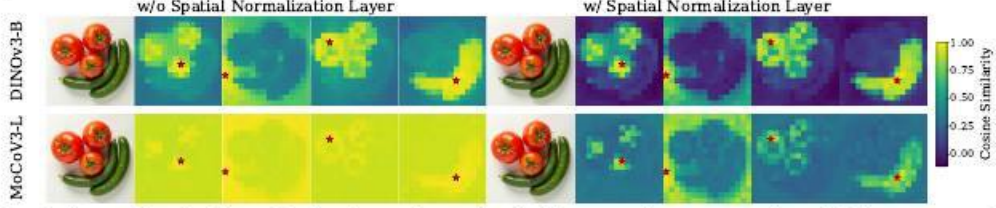
Spatial normalization layer. Similar to Siméoni et al. (2025), we find that patch tokens from pretrained vision encoders often contain a substantial global component. The high linear probing scores for the mean of patch tokens provide one indication of this behavior (Figure 14). Figure 6b further shows that mixing this global component (the mean of patch tokens) into local patch tokens can improve global performance while reducing spatial contrast between individual tokens. As a result, tokens from different image regions, such as foreground objects and background, can become overly similar.

Given the results in §3, we hypothesize that partially suppressing this global component can improve the spatial contrast among patch tokens. Higher contrast should provide a clearer spatial alignment signal, expressed through pairwise similarity between patch tokens, and therefore improve REPA performance. To implement this idea, we add a simple spatial normalization layer (Ulyanov et al., 2016) to the patch tokens of the target representation:

$$\mathbf{y} = \frac{\mathbf{x} - \gamma \mathbb{E}[\mathbf{x}]}{\sqrt{\text{Var}[\mathbf{x}] + \epsilon}},$$



(a) *Simpler spatial projection layer for REPA.* Standard MLP projection layer in REPA (middle) loses spatial information while transferring features from target representation (left) to diffusion features. Instead using a simpler convolution layer leads to better spatial information transfer (right).



(b) *Spatial normalization layer.* Patch tokens of pretrained vision encoders can contain a global component that limits spatial contrast. This makes tokens from one semantic region (e.g., tomato) similar to unrelated regions (e.g., background or cucumber). We therefore reduce part of this global component to increase spatial contrast between patch tokens and provide a clearer alignment signal for generation.



(c) *Overall impact of iREPA on the spatial structure of diffusion features.*

Figure 6: Motivation for two simple iREPA modifications that strengthen spatial feature transfer from the target representation to diffusion features. All results are reported with SiTL/2 and REPA at 100K.

where $\mathbf{x} \in \mathbb{R}^{B \times T \times D}$ represents the patch tokens, the expectation and variance are computed across the spatial dimension, and $\epsilon = 10^{-6}$ is used for numerical stability.

Algorithm 1 Summary of key iREPA modifications to standard REPA training

```
# 1. Conv projection instead of MLP
proj_layer = nn.Conv2d(D_in, D_out, kernel_size=3, padding=1)

# 2. Spatial normalization on encoder features [B, T, D]
x = x - gamma * x.mean(dim=1, keepdim=True)
x = x / (x.std(dim=1, keepdim=True) + 1e-6)
```

4.1 EXPERIMENTS

We evaluate whether this spatially focused recipe improves REPA under the same ImageNet 256×256 experimental setting used throughout the paper. The experiments ask whether the same design principle transfers across target representations, model scales, alignment depths, and training variants:

- Does iREPA improve diffusion-transformer convergence over REPA across diverse external representations? (Figure 7, 8, 12, 13, 15)
- Does iREPA scale with model size and remain effective under variations in alignment depth and training settings? (Table 1 [a,b,c], 2, 4, 5, 6, and, Figure 12, 16)
- Does the spatial-transfer principle extend to recent alignment and diffusion variants such as REPA-E (Leng et al., 2025a), MeanFlow w/ REPA (Geng et al., 2025), and pixel-space diffusion with JiT w/ REPA (Li & He, 2025b)? (Table 3 [a,b]; JiT w/ REPA (Li & He, 2025b))

Convergence Speed. We evaluate iREPA convergence across diverse vision encoders (DINOv3-B, WebSSL-1B, PE-Core-G, CLIP-L, MoCov3, PE-Lang-G) and model sizes (SiT-XL/2 and SiT-B/2). Fig. 7 shows that iREPA consistently reaches better FID faster than baseline REPA across both target-representation and model-size variations.

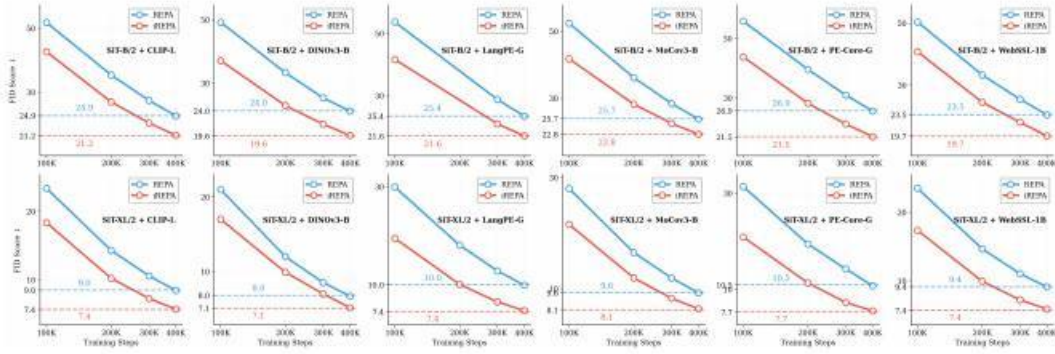


Figure 7: Accentuating spatial features helps consistently improve convergence speed.

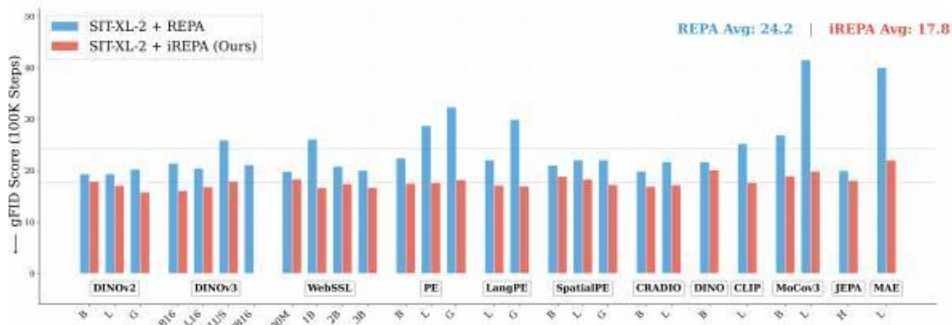


Figure 8: **Variation in target representation.** Across all 27 vision encoders, we find that accentuating transfer of spatial features from target representation to diffusion features (iREPA) leads to consistent improvements in convergence speed. See Appendix C for more results across diverse settings.

Target representation. We analyze transfer across vision encoders in Fig. 8 and Table 4. iREPA improves generation quality for every encoder included in this comparison, and Appendix C, E provides the corresponding comparisons across all 27 encoders.

Enc. Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$	Model Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$	Aln. Depth	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$
PE-B (90M)	59.6	22.5	6.23	0.63	0.60	SiT-B	27.5	49.50	7.00	0.46	0.59	Layer 4	28.1	49.0	6.83	0.46	0.60
+iREPA	73.3	17.5	6.04	0.66	0.62	+iREPA	34.1	43.37	7.87	0.50	0.60	+iREPA	41.7	36.0	6.61	0.52	0.62
PE-L (320M)	48.6	28.7	6.53	0.59	0.60	SiT-L	55.7	24.10	6.25	0.62	0.60	Layer 6	26.9	50.6	7.00	0.46	0.59
+iREPA	76.3	17.6	6.36	0.65	0.61	+iREPA	66.9	20.28	6.14	0.63	0.62	+iREPA	42.4	35.3	6.90	0.53	0.62
PE-G (1.88B)	42.7	32.3	6.70	0.57	0.59	SiT-XL	70.3	19.06	5.83	0.65	0.61	Layer 8	24.7	54.8	7.35	0.44	0.58
+iREPA	70.8	19.5	6.35	0.64	0.61	+iREPA	77.9	16.96	6.26	0.66	0.61	+iREPA	38.3	38.9	7.19	0.51	0.62

(a) Vision Encoder size

(b) Model size

(c) Encoder depth (SiT-B/2)

Table 1: **Variation in training settings.** iREPA provides consistent gains over baseline REPA across the evaluated training settings. Unless otherwise specified, all results use Dinov2 as the encoder, SiT-XL/2, 100K steps, and vanilla REPA as the baseline.

Encoder size. We next analyze whether iREPA remains effective across encoder sizes. Table 1a compares PE-B (90M), PE-L (320M), and PE-G (1.88B). iREPA improves performance for all encoder sizes, and the relative improvement increases with encoder size (22.2% for PE-B, 38.8% for PE-L, 39.6% for PE-G).

Scalability. Table 1b evaluates iREPA across generative model scales. The spatial modifications improve performance at each scale, with larger percentage gains for larger models, indicating that the benefits of spatial transfer scale with model capacity in these experiments.

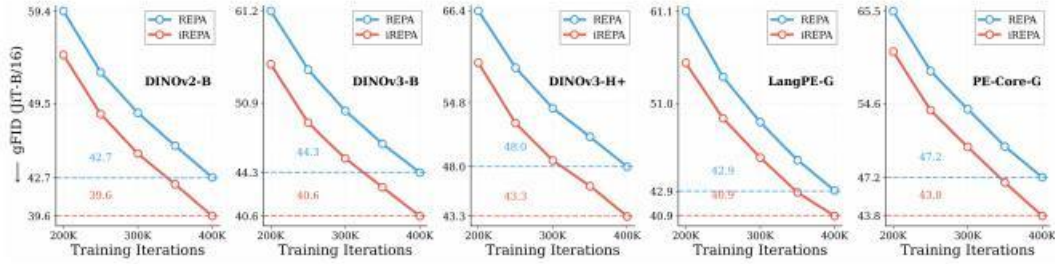


Figure 9: **Convergence comparison with pixel-space diffusion (JiT).** FID convergence curves comparing REPA and iREPA with pixel-space JiT (Li & He, 2025b) across different vision encoders. Accentuating spatial-structure transfer improves convergence speed across different vision encoders for pixel-space diffusion models. All results are reported with JiTB/16 (Li & He, 2025b), 256 batch size and without classifier-free guidance. Refer Table 10 for further results.

Method	DiNOv2-B			DiNOv3-B			WebSSL-1B			PE-Core-G		
	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓
Baseline REPA	19.06	70.3	5.83	21.47	63.4	6.19	26.10	53.0	6.90	32.35	42.7	6.70
iREPA (w/o spatial norm)	18.52	73.3	6.11	17.76	74.7	5.81	21.17	64.6	6.27	24.97	57.4	6.21
iREPA (w/o conv proj)	17.66	72.8	6.03	18.28	70.8	6.18	18.44	71.0	6.22	21.72	61.5	6.26
iREPA (full)	16.96	77.9	6.26	16.26	78.8	6.14	16.66	77.5	6.18	18.19	75.0	6.03

Table 2: **Ablation on different components.** Across different encoders, both the spatial normalization layer and the convolution projection layer (§4) improve convergence speed, with the best results obtained by using both components. All results reported at 100K steps and SiT-XL/2.

Encoder depth. Table 1c tests different alignment depths using SiT-B/2 at 100K iterations with DiNOv3-B as the target representation. iREPA improves over baseline REPA at each evaluated depth, suggesting that the spatial recipe is not tied to a single alignment layer.

Ablation on different components. We isolate the two iREPA components in Table 2. Both spatial normalization and the convolution projection layer improve generation quality relative to baseline REPA, and combining them gives the best result.

Training recipe. Finally, we test whether the same spatial modifications remain useful beyond the base REPA recipe by evaluating REPA-E (Leng et al., 2025a) and MeanFlow w/ REPA (Geng et al., 2025). Table 3 shows convergence gains across these training recipes.

Classifier-free guidance. We evaluate iREPA with classifier-free guidance in Table 4, which reports results across vision encoders at 400K training iterations. Across these encoders, iREPA improves convergence both with and without classifier-free guidance.

Pixel-space diffusion (JiT). We also evaluate iREPA on pixel-space diffusion models such as JiT-B (Li & He, 2025b). Figure 9 compares REPA and iREPA in the JiT setting and shows that emphasizing spatial transfer accelerates convergence across multiple vision encoders (e.g., DiNOv2, DiNOv3, PE etc.).

- **Finding 3.** Accentuating transfer of spatial structure from target representation to diffusion features helps boost convergence speed with representation alignment (REPA).

5 RELATED WORK

We focus here on the related work most directly connected to representation alignment and spatially structured visual representations; Appendix H provides a broader discussion.

Representation alignment for generation. A growing body of work uses external visual representations to improve diffusion model training (Pernias et al., 2023; Fuest et al., 2024). Recent methods (Yu et al., 2024; Yao & Wang, 2025; Leng et al., 2025a,b; Kouzelis et al., 2025) show that aligning internal diffusion features with clean-image features from pretrained vision encoders can substantially

Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
WebSSL-1B	52.8	26.5	5.20	0.620	0.585
+iREPA-E	87.0	13.2	5.28	0.699	0.598
PE-G	50.9	25.9	5.68	0.612	0.576
+iREPA-E	80.0	16.4	5.40	0.667	0.616
DINOv3-B	82.2	14.4	4.68	0.694	0.596
+iREPA-E	93.6	11.7	4.57	0.703	0.613
DINOv2-B	87.5	12.9	5.40	0.708	0.586
+iREPA-E	91.3	12.1	4.86	0.712	0.602

(a) REPA-E (SiT-XL/2)

Encoder	w/o CFG				w/ CFG			
	4 NFE		1 NFE		4 NFE		1 NFE	
	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$
WebSSL-1B	27.22	51.40	24.14	58.67	87.85	16.59	69.06	23.74
+iREPA	31.48	45.67	27.33	55.70	100.74	13.89	78.67	20.69
DINOv3-B	28.36	49.58	25.47	57.01	93.30	15.56	72.41	22.55
+iREPA	33.63	44.52	29.67	53.75	124.54	11.05	98.93	17.32
DINOv2-B	28.77	48.87	25.43	56.63	94.40	15.28	71.95	22.17
+iREPA	33.63	44.52	29.67	53.75	111.44	12.59	90.25	18.62

(b) MeanFlow w/ REPA (SiT-B/2)

Table 3: **Variation in training recipes.** We apply the spatial iREPA modifications on top of REPA-E (a) and Meanflow with REPA (b), obtaining consistent gains in both settings. For Meanflow, we report results at 1 and 4 NFEs. CFG value of 2.0 is used for Meanflow. All results are reported at 100K training iterations.

Vision Encoder	Steps	w/o CFG					w/ CFG				
		IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	100K	69.20	19.3	5.89	0.64	0.61	157.2	6.35	5.91	0.769	0.536
+iREPA	100K	77.92	16.9	6.26	0.66	0.61	179.3	5.15	6.23	0.783	0.544
DINOv2-B	400K	127.4	7.76	5.06	0.70	0.66	263.0	1.98	4.60	0.799	0.610
+iREPA	400K	128.6	7.52	4.89	0.71	0.65	268.8	1.93	4.59	0.799	0.600
DINOv3-B	100K	63.64	21.4	6.14	0.63	0.60	144.0	7.57	6.09	0.762	0.526
+iREPA	100K	78.79	16.2	6.14	0.66	0.61	181.9	4.87	6.10	0.780	0.547
DINOv3-B	400K	126.7	8.10	5.06	0.70	0.66	261.2	1.99	4.58	0.799	0.609
+iREPA	400K	132.9	7.13	4.93	0.71	0.66	272.4	1.89	4.58	0.799	0.600
WebSSL-1B	100K	53.87	25.5	6.57	0.61	0.59	124.0	9.59	6.37	0.756	0.515
+iREPA	100K	77.47	16.6	6.18	0.66	0.61	177.6	5.09	6.11	0.787	0.538
WebSSL-1B	400K	116.9	9.39	5.14	0.70	0.64	250.5	2.24	4.61	0.809	0.580
+iREPA	400K	130.8	7.48	4.91	0.70	0.65	271.6	1.90	4.58	0.798	0.609
PE-Core-G	100K	42.74	32.3	6.70	0.57	0.59	97.2	14.1	6.56	0.714	0.525
+iREPA	100K	75.01	18.1	6.03	0.64	0.61	176.8	5.66	6.08	0.771	0.544
PE-Core-G	400K	109.4	10.4	5.00	0.69	0.64	238.4	2.44	4.57	0.805	0.585
+iREPA	400K	132.0	7.78	5.02	0.70	0.65	275.4	1.93	4.59	0.796	0.606

Table 4: **Results across encoders with and without classifier-free guidance.** SiT-XL/2 is used as the base model. See Appendix E for detailed SiT-B/2 and SiT-L/2 results across all encoders.

improve training efficiency and generation quality. This idea has also been extended beyond image generation, including video generation (Zhang et al., 2025), 3D generation (Wu et al., 2025), and unified generative models (Ma et al., 2025a). Despite these empirical successes across tasks and model families, the field has less clarity on which properties of the external representation are most responsible for the gains. Our work addresses this question by separating global semantic information from spatial structure and using the resulting insight to design an improved training recipe.

Spatial vs global information tradeoff in pretrained vision encoders. Recent studies analyze the tradeoff between global and spatial information in pretrained vision encoders (Bolya et al., 2025; Siméoni et al., 2025). Siméoni et al. (2025) show that continued training of self-supervised vision representations can increase similarity between the global CLS token and patch tokens, which can hurt dense spatial tasks. Other work explicitly trains spatially tuned models for dense prediction settings (Bolya et al., 2025; Heinrich et al., 2024). We study this tradeoff in the context of generative representation alignment and show that, within our evaluated settings, the spatial structure of a vision encoder is more predictive of generation performance than its global information. This suggests that future external representations for generation should be selected and trained with spatial alignment signals in mind.

6 CONCLUSION

We studied a general question in representation alignment for generative models: when a pretrained vision encoder is used as an external target, what property of that representation most improves diffusion training? Across a large empirical analysis, we find that spatial structure, rather than global information measured by ImageNet-1K accuracy, is the stronger predictor of REPA effectiveness. We

then test this insight through two simple modifications that emphasize spatial transfer from the target representation to diffusion features. The resulting method, iREPA, improves REPA convergence across the evaluated range of vision encoders, model sizes, and training recipes. More broadly, our results suggest that the design and selection of alignment targets for generation should account for spatial organization, not only global semantic accuracy.

REPRODUCIBILITY STATEMENT

We report the implementation details and hyperparameters needed to reproduce the evaluated settings in Appendix G. Our code, model checkpoints, and analysis results will be released upon acceptance.

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A DETAILED CORRELATION ANALYSIS

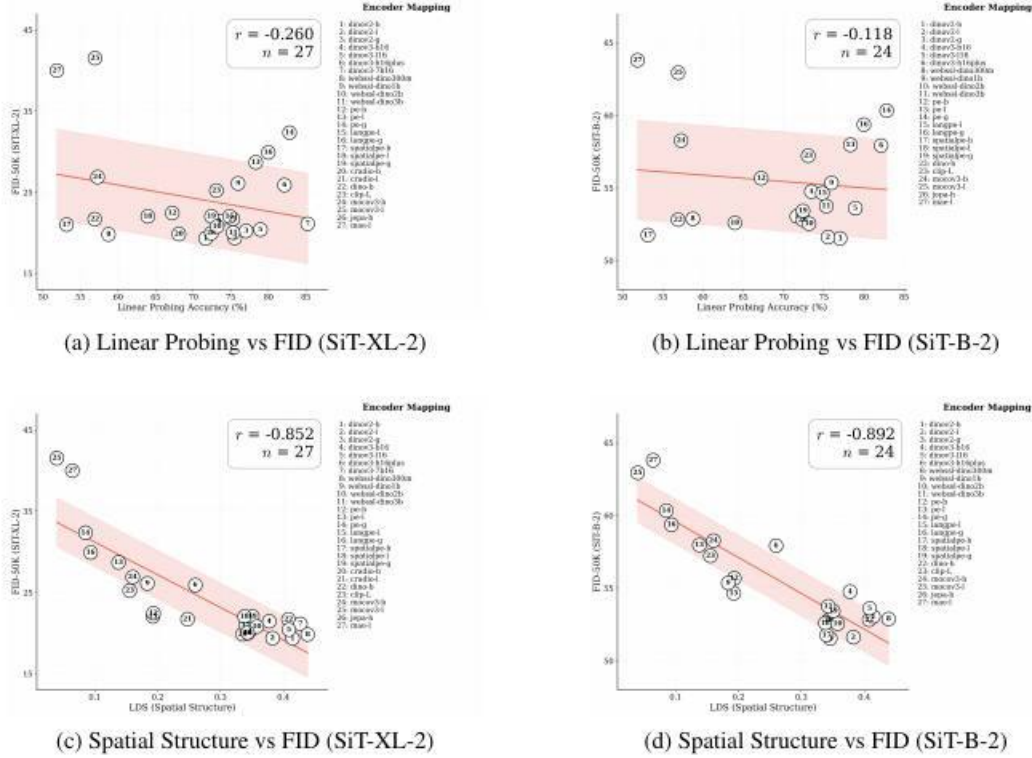


Figure 10: Detailed correlation analysis with encoder labels. The top row shows the correlation between linear probing accuracy and FID for SiT-XL-2 and SiT-B-2. The bottom row shows the corresponding correlation between spatial structure (LDS) and FID. Each point is labeled with its encoder number (see legend). Across both model scales, the spatial structure metric correlates more strongly with generation quality than linear probing.

B SPATIAL SELF-SIMILARITY METRICS



Figure 11: **Explaining semantic region self-similarity (SRSS) metric.** Visual explanation of SRSS metric: For each image, we compute a mask using SAM2 and select anchor-positive-negative triplets. The SSM measures the difference in cosine similarity between anchor-positive pairs (within mask) and anchor-negative pairs (outside mask). Intuitively, larger values indicate that patches on the same semantic region are more similar than patches on unrelated regions — indicating better spatial structure preservation.

Setup. Given an image $\mathcal{I}$ and a vision encoder $\mathcal{E}$, we extract patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ with $T = H \times W$ and indices $t \in [T]$ placed on an $H \times W$ lattice. We use the Manhattan distance $d : [T] \times [T] \rightarrow \mathbb{N}$ to describe token separation, and define the cosine self-similarity kernel as

$$K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2} \in [-1, 1],$$

which is a standard local self-similarity measure (Shechtman & Irani, 2007).

- *Patch token representation:* $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ where $T = H \times W$ is the spatial grid of patches (patch index set $[T]$). Also let $d : [T] \times [T] \rightarrow \mathbb{N}$ be Manhattan distance on the $H \times W$ grid.
- *Self-similarity kernel:* $K : [T] \times [T] \rightarrow \mathbb{R}$ measuring self-similarity (Shechtman & Irani, 2007) between patch tokens. We use the cosine kernel $K_{\mathcal{X}}(t, t') = \langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle / (\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2)$.
- *Spatial self-similarity metric:* a functional $m : \mathcal{K} \times \mathcal{D} \rightarrow \mathbb{R}$ measures how self-similarity K between patch tokens varies with lattice distance d . Larger values indicate stronger spatial organization, meaning nearby patches are more similar to each other than distant patches.

We next describe several simple metrics that instantiate this general notion of spatial self-similarity for target representations.

Local vs. Distant Similarity (LDS). We first consider a simple correlogram contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') < r_{\text{near}}] - \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') \geq r_{\text{far}}],$$

where r_{near} and r_{far} are hyperparameters. By default, we use $r_{\text{near}} = r_{\text{far}} = H/2$, and the observed correlations are robust to their exact values. Larger LDS values indicate that, *on average*, nearby patches are more similar than distant patches.

Correlation Decay Slope (CDS). Given the patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$, we compute the spatial correlogram $g_{\mathcal{X}}(\delta) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') = \delta]$ for distances $\delta \in \{1, \dots, \Delta\}$. We then fit a least-squares line $\hat{g}_{\mathcal{X}}(\delta) \approx \alpha + \beta\delta$ and define:

$$\text{CDS}(\mathcal{X}) = -\hat{\beta}$$

where $\hat{\beta}$ is the fitted slope. Larger CDS values indicate that similarity decays faster with distance, which corresponds to stronger spatial organization.

Semantic-Region Self-Similarity (SRSS). For each image, we sample a binary mask $\mathbf{M} \in \mathbb{R}^{H \times W}$ containing a random object using SAM2 (Ravi et al., 2024) and downsample it to $H \times W$. We then choose anchor, positive, and negative points. The anchor and positive points are sampled from within the mask and should be semantically related, whereas the negative point is sampled outside the mask and should be less related to the anchor. If the encoder preserves spatial structure, anchor-positive cosine similarity should exceed anchor-negative cosine similarity. We therefore define the Spatial Structure Metric (SSM) as:

$$\text{SSM}(\mathbf{P}) = \mathbb{E}_{\text{anchor} \in \mathbf{M}} [\cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{pos}}) - \cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{neg}})]$$

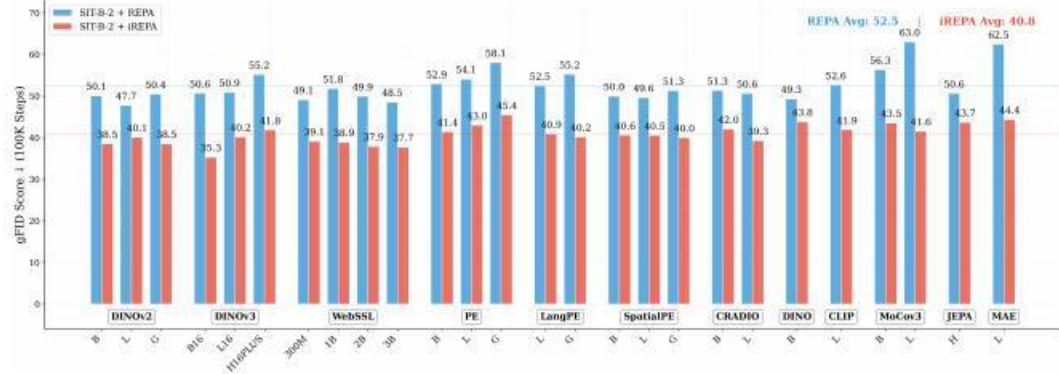
where positive points lie within Manhattan distance $d \leq r_{near}$ from the anchor inside the mask, and negative points lie at distance $d \geq r_{far}$ outside the mask.

RMS Spatial Contrast (RMSC). Finally, we use a simple contrast metric to measure the spatial diversity of patch token representations. Given normalized features $\hat{\mathbf{x}}_t = \mathbf{x}_t / \|\mathbf{x}_t\|_2$ for each patch t , we compute:

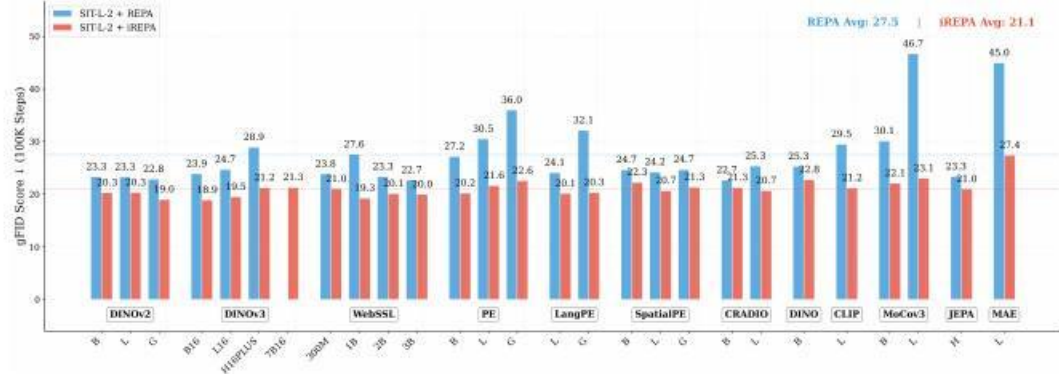
$$\text{RMSC}(\mathcal{X}) = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{x}}_t - \bar{\mathbf{x}}\|_2^2}$$

where $\bar{\mathbf{x}} = \frac{1}{T} \sum_{t=1}^T \hat{\mathbf{x}}_t$ is the mean of normalized features across all spatial locations. Higher RMSC values indicate greater spatial diversity in the feature representations and therefore more preserved spatial structure. Lower values indicate more uniform feature distributions, consistent with spatial structure being diminished by global information (Siméoni et al., 2025).

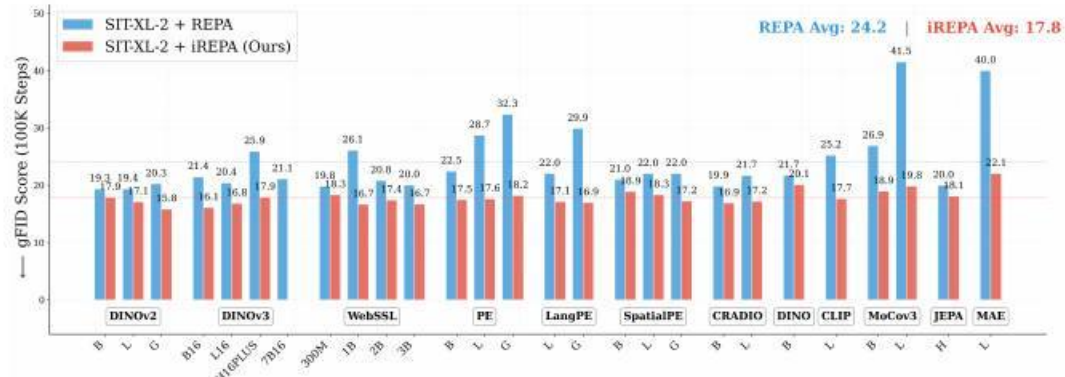
C VISION ENCODER VARIATION



(a) SiT-B-2 (130M parameters)



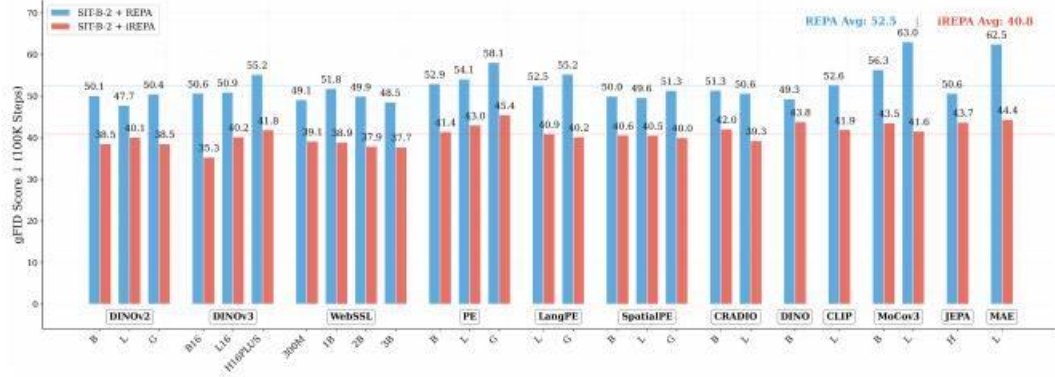
(b) SiT-L-2 (458M parameters)



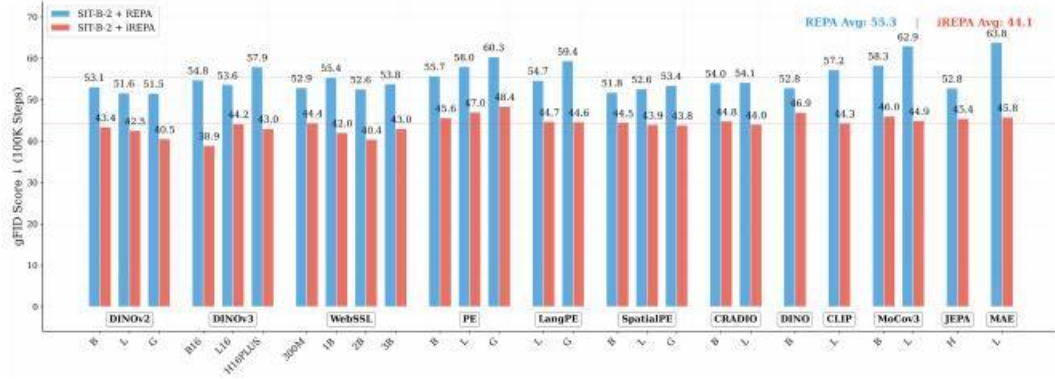
(c) SiT-XL-2 (675M parameters)

Figure 12: **Variation in vision encoders.** Ablation studies showing iREPA improvements across different vision encoders for SiT-B-2, SiT-L-2, and SiT-XL-2 models. iREPA consistently improves generation quality across all encoders and model sizes.

D ENCODER DEPTH VARIATION



(a) SiT-B-2 with 6 encoder layers



(b) SiT-B-2 with 8 encoder layers

Figure 13: **Effect of encoder depth on iREPA performance.** Comparison of SiT-B-2 performance with different encoder depths (6 vs 8 layers) across vision encoders. iREPA improves performance for both encoder depths, with slightly better performance observed with 8 encoder layers. Both configurations show FID improvements when the spatial modifications are applied.

E COMPREHENSIVE RESULTS ACROSS VISION ENCODERS

E.1 SiT-B/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	59.44	24.84	6.46	0.584	0.645
+iREPA	69.70	21.26	6.64	0.601	0.644
DINOv2-B	64.92	22.75	6.54	0.593	0.653
+iREPA	70.88	21.40	6.77	0.597	0.653
DINOv2-L	66.65	22.46	6.53	0.594	0.650
+iREPA	72.41	20.99	6.84	0.600	0.650
DINOv3-B16	62.57	23.91	6.77	0.584	0.647
+iREPA	75.48	19.65	6.68	0.606	0.649
DINOv3-H16+	61.91	24.82	6.57	0.579	0.645
+iREPA	71.76	21.80	6.84	0.592	0.655
DINO-B	58.37	24.71	6.32	0.586	0.635
+iREPA	57.55	24.41	6.43	0.587	0.638
LangPE-G	58.42	25.37	6.32	0.583	0.641
+iREPA	66.88	21.62	6.65	0.604	0.649
MoCov3-B	57.61	25.60	6.25	0.579	0.640
+iREPA	61.41	22.80	6.46	0.596	0.639
PE-B	58.86	25.78	6.44	0.58	0.64
+iREPA	69.50	21.15	6.66	0.601	0.659
PE-G	56.09	26.84	6.45	0.568	0.641
+iREPA	70.43	21.52	6.58	0.598	0.651
SpatialPE-B	59.19	24.28	6.37	0.586	0.646
+iREPA	61.50	22.71	6.41	0.604	0.644
SpatialPE-L	62.13	23.87	6.51	0.59	0.65
+iREPA	68.83	21.07	6.59	0.607	0.651
WebSSL-1B	63.20	23.48	6.33	0.593	0.646
+iREPA	73.54	19.74	6.63	0.612	0.646
WebSSL-2B	66.56	22.27	6.69	0.596	0.654
+iREPA	72.13	20.47	6.89	0.609	0.648
WebSSL-300M	62.06	23.64	6.48	0.590	0.646
+iREPA	66.83	21.46	6.71	0.606	0.653
WebSSL-3B	65.91	22.55	6.53	0.598	0.654
+iREPA	71.98	20.64	6.71	0.607	0.650

Table 5: **SiT-B/2 performance across vision encoders at 400K iterations.** iREPA consistently improves generation quality. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

E.2 SiT-L/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	107.7	10.6	5.20	0.682	0.643
+iREPA	115.8	9.41	5.15	0.689	0.653
DINOv2-B	113.3	9.61	5.11	0.684	0.653
+iREPA	113.9	9.36	5.11	0.690	0.655
DINOv3-B16	115.3	9.41	5.24	0.687	0.651
+iREPA	118.1	9.06	5.25	0.685	0.650
DINOv3-H16+	110.7	10.6	5.32	0.677	0.652
+iREPA	120.5	9.64	5.29	0.681	0.663
LangPE-G	102.8	11.2	5.13	0.684	0.637
+iREPA	112.7	9.37	5.02	0.690	0.653
MoCov3-B	102.1	11.0	5.14	0.685	0.635
+iREPA	104.7	10.1	5.00	0.690	0.641
PE-G	98.38	12.6	5.31	0.670	0.646
+iREPA	117.1	9.65	5.33	0.683	0.651
WebSSL-1B	107.9	10.5	5.13	0.686	0.644
+iREPA	121.5	8.62	5.01	0.696	0.653

Table 6: **SiT-L/2 performance across vision encoders at 400K iterations.** iREPA consistently improves generation quality across all encoders, with particularly strong gains in FID and IS metrics. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

F ADDITIONAL EXPERIMENTAL RESULTS

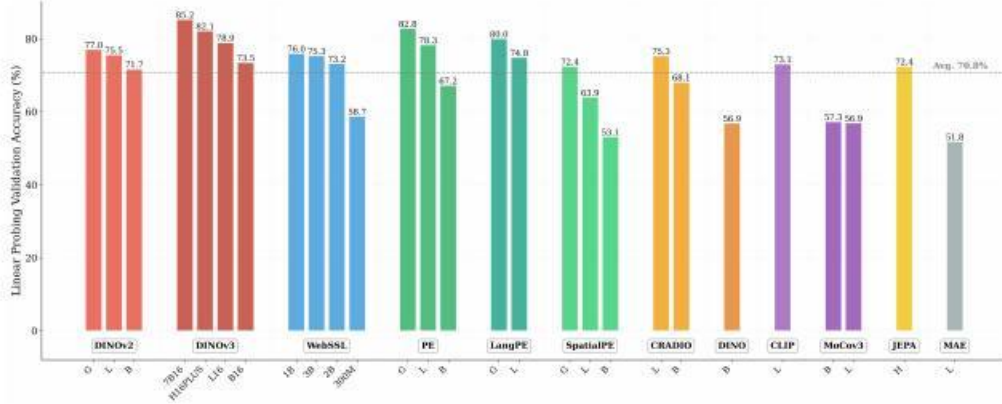


Figure 14: **Global information in mean patch tokens.** In addition to the [CLS] token, the mean of patch tokens contains substantial global semantic information. Although this component can improve global performance, it reduces contrast between individual patch tokens and may hinder spatial structure transfer. We therefore remove part of this mean-token global information to improve spatial transfer during representation alignment (§4).

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	262.97	1.98	4.60	0.799	0.610
+iREPA (Ours)	268.79	1.93	4.59	0.799	0.600
DINOv3-B	261.18	1.99	4.58	0.799	0.609
+iREPA (Ours)	272.41	1.89	4.58	0.799	0.600
WebSSL-1B	250.53	2.24	4.61	0.809	0.580
+iREPA (Ours)	271.59	1.90	4.58	0.798	0.609
PE-G	238.37	2.44	4.57	0.805	0.585
+iREPA (Ours)	275.36	1.93	4.59	0.796	0.606

Table 7: **Generation quality with classifier-free guidance.** Comparison of REPA and iREPA with CFG (scale 2.0) across different vision encoders. All experiments use SiT-XL/2 trained for 400K iterations with 250 sampling steps. iREPA improves both IS and FID across the evaluated encoders.

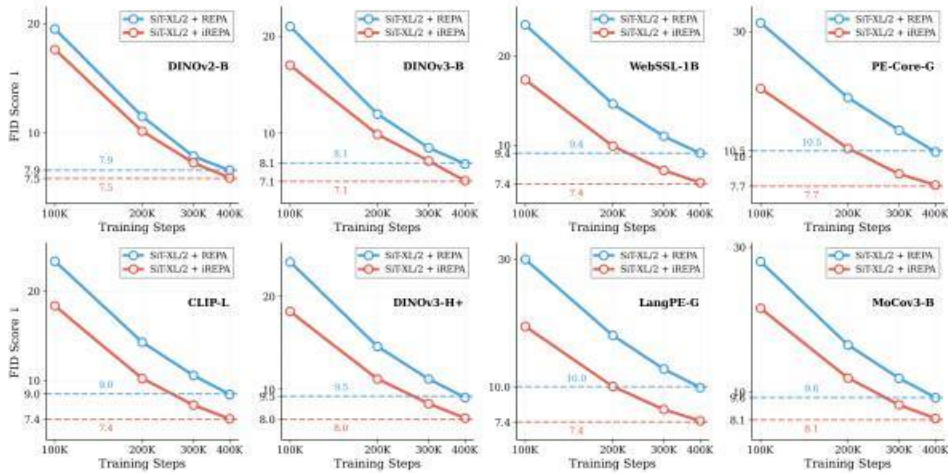


Figure 15: **Accentuating spatial features helps consistently improve convergence speed.** Results for SiT-XL/2 with REPA and iREPA.

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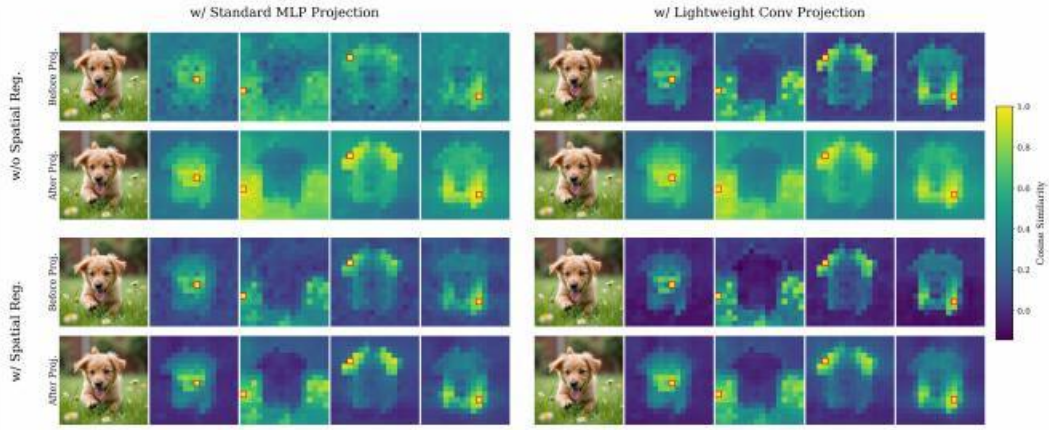


Figure 16: **Visualizing the impact of two straightforward improvements to enhance spatial feature transfer.** **First**, we find that standard MLP projection layer (top-left) losses spatial information while transferring features from the pretrained encoder (after projection) to diffusion model (before projection). Instead using a simpler convolution layer better preserves the spatial information transfer (top-right). **Second**, we observe that vision encoder features often have limited spatial contrast (§2. This causes the tokens in one semantic region (*e.g.*, dog) to show quite decent cosine similarity with unrelated tokens (*e.g.*, background). We address this with a simple spatial regularization layer, which accentuates the spatial contrast in the learned representation — leading to better generation performance. Best results are obtained while using both (refer Table 2).

G IMPLEMENTATION DETAILS

Training setup. We follow the same setup as in REPA (Yu et al., 2024) unless otherwise specified. All training is conducted on the ImageNet (Deng et al., 2009) training split. For preprocessing, we adopt the protocol from ADM (Dhariwal & Nichol, 2021), where each image is center-cropped and resized to 256×256 resolution. We use `stabilityai/sd-vae-ft-mse` VAE (AI, n.d.) throughout our diffusion training and inference. For spatial normalization layer we use $\gamma \in [0.6, 0.8]$ and for projection layer we use a convolutional layer with kernel size 3 and padding 1. For optimization, we use AdamW (Kingma & Ba, 2014; Loshchilov, 2017) with a constant learning rate of 1×10^{-4} , and a global batch size of 256. During training, we use `bfloat16` mixed precision and `torch.compile` to accelerate training, and gradient clipping and exponential moving average (EMA) to the generative model for stable optimization.

For REPA-E (Leng et al., 2025a) and JiT (Li & He, 2025a) experiments, we use the official open-source implementation. For MeanFlow (Geng et al., 2025) experiments, we received the implementation (including REPA) from original authors and adapt it to introduce the two straightforward changes for iREPA (§4).

Evaluation. For image generation evaluation, we strictly follow the ADM setup (Dhariwal & Nichol, 2021). We report generation quality using Fréchet inception distance (gFID) (Heusel et al., 2017), structural FID (sFID) (Nash et al., 2021), inception score (IS) (Salimans et al., 2016), precision (Prec.) and recall (Rec.) (Kynkäänniemi et al., 2019), measured on 50K generated images. For sampling, we follow the approach in REPA (Yu et al., 2024), using the SDE Euler-Maruyama sampler with 250 steps. For JiT (Li & He, 2025a), we use 50 inference steps following official implementation.

H MORE DISCUSSION ON RELATED WORK

Classical spatial features in computer vision. Spatial feature extraction has long been fundamental to computer vision. Classical methods like SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), SURF (Bay et al., 2006), and ORB (Rublee et al., 2011) providing robust local descriptors for tasks ranging from object detection to image matching. While these handcrafted features excel at capturing local spatial patterns and geometric invariances, recent work in generative modeling has primarily focused on representations from modern self-supervised methods that demonstrate strong global classification performance, such as DINOv2 (Oquab et al., 2024) and CLIP (Radford et al., 2021). Our findings suggest a different perspective: since spatial structure preservation is critical for generation quality, even classical spatial features could potentially improve diffusion training when properly aligned. This highlights the potential of leveraging the full spectrum of spatial feature extractors, from traditional handcrafted descriptors to modern learned representations, provided that they maintain strong spatial coherence.

Pretrained visual encoders for generative models. Pretrained visual encoders have supported generative models in several capacities: as discriminators to accelerate GAN convergence (Goodfellow et al., 2020; Sauer et al., 2021; 2022; 2023; Radford et al., 2021), as teachers in adversarial distillation for diffusion models (Sauer et al., 2024; Kang et al., 2024), and more recently as alignment targets. In GANs (Goodfellow et al., 2020), pretrained features have not only improved convergence speed but also enabled scaling to large datasets, as demonstrated by StyleGAN-XL (Sauer et al., 2022) and StyleGAN-T (Sauer et al., 2023) with CLIP (Radford et al., 2021) features. For diffusion models, adversarial distillation leverages pretrained encoders to guide student networks toward higher-fidelity samples, showing clear improvements in perceptual quality. In particular, REPA (Yu et al., 2024) aligns diffusion features with external encoders, demonstrating that representation alignment can improve both generation convergence and quality. Building on this direction, we focus not only on alignment but specifically on spatial structure perseverance rather than their discriminative capabilities.

Denosing transformers. Transformer architectures have become the dominant backbone for scalable generative modeling, with various formulations including diffusion transformers (Peebles & Xie, 2023) and flow matching variants (Ma et al., 2024). Recent architectures like GenTron (Chen et al., 2023) scale transformers to over 3B parameters for text-to-image synthesis. U-ViT (Bao et al., 2023) demonstrates that pure transformer backbones without U-Net inductive biases can achieve competitive performance. ARDiT (Tang et al., 2024) and DART (Jing et al., 2024) explore

autoregressive formulations that combine denoising with sequential generation, enabling flexible trade-offs between quality and speed. Despite these architectural advances, training these models from scratch remains computationally expensive, often requiring millions of iterations to achieve good generation quality. While representation alignment methods like REPA (Yu et al., 2024) have shown that pretrained features can dramatically accelerate convergence, the mechanisms behind this improvement remained unclear. Our analysis reveals that the key benefit comes from preserving spatial structure rather than semantic alignment, explaining why certain encoders provide stronger acceleration than others and guiding the design of more effective alignment strategies. While (Wang et al., 2025c) propose improvements to REPA training through early stopping, they hypothesize that “REPA predominantly distills global semantic information while leaving structural information untapped.” In contrast, our analysis reveals that spatial structure (not global semantic information) already plays a very significant role in the effectiveness of REPA (§2, §3). Thus, while we indeed find that spatial structure remains underexploited (§4) in REPA, we surprisingly find that majority of improvements in REPA are already coming from introducing an inductive bias for spatial structure (not global information).

Denoising as self-supervised learning task. The connection between denoising and representation learning has been explored from multiple perspectives. Early work by Abstreiter et al. (2021) extended diffusion objectives for representation learning, demonstrating that denoising naturally learns meaningful features. SODA (Hudson et al., 2024) introduces a diffusion model with an information bottleneck to learn compact representations. Chen et al. (2024) deconstruct diffusion models and find that a simple denoising autoencoder suffices for strong self-supervised performance. Wang & He (2025) introduce a dispersive loss to encourage internal representation diversity in diffusion models, improving generative performance without external encoders. Similarly, Jiang et al. (2025) propose Self-Representation Alignment (SRA) to align a diffusion transformer’s latent features across noise levels, providing self-guidance without an auxiliary model. These works establish denoising as a fundamental self-supervised task that naturally encourages learning of robust features. Our findings complement this view by showing that when diffusion models are aligned with strong spatial representations from self-supervised encoders, both the generative and discriminative capabilities improve, suggesting that spatial structure preservation is a key factor in this synergy.

Scaling self-supervised vision encoders. Recent years have seen remarkable progress in scaling self-supervised vision models to unprecedented sizes and datasets. DINOv3 (Siméoni et al., 2025) trains a 7B parameter ViT on 1.7 billion images without labels by aligning representations from different augmentations. WebSSL (Fan et al., 2025) demonstrates that visual models trained on more than 2 billion images can match language-supervised models like CLIP (Radford et al., 2021) on vision-language tasks without language supervision. C-RADIO (Heinrich et al., 2024) combines multiple teacher models through distillation to create versatile encoders that excel across diverse visual domains. I-JEPA (Assran et al., 2023) explores predictive architectures that learn by predicting masked regions in representation space, instead of reconstructing pixels directly. SAM (Ravi et al., 2024) specializes in promptable segmentation through large-scale supervised training. The Perception Encoder family (Bolya et al., 2025; Cho et al., 2025) shows that intermediate features often outperform final representations for dense prediction tasks. While these models are typically evaluated on global tasks like image classification, our work reveals a crucial insight: strong performance on discriminative benchmarks does not necessarily translate to better generation quality. Instead, we find that encoders preserving spatial structure, regardless of their classification accuracy—provide the most benefit for diffusion training. This suggests that the evaluation metrics for self-supervised encoders should be reconsidered when targeting generative applications, with spatial coherence being as important as semantic understanding.

I NOTE ON LLM USAGE

We use GPT-5 (OpenAI, 2025) for considering different variations of the spatial structure metrics discussed in the paper §3. Furthermore, all figures in the paper are directly generated from our experiment logs and checkpoints using Claude-Code (Anthropic, 2025). Additionally we use LLM help for searching and formulating relevant work in Appendix H. We use cursor in some parts to help with paper writing.

J ADDITIONAL DISCUSSION ON SPATIAL STRUCTURE METRICS

Spatial structure metrics can be used to measure the representation gap. Yu et al. (2024) use linear probing to measure the representation gap, interpreting increased validation accuracy as evidence for REPA effectiveness. We find that spatial structure metrics (SSM) provide a complementary and more spatially grounded view of this gap across models. As shown in Figure 17, representation alignment helps close the gap between the SSM of a pretrained encoder such as DINOv2 and the SSM of diffusion features.

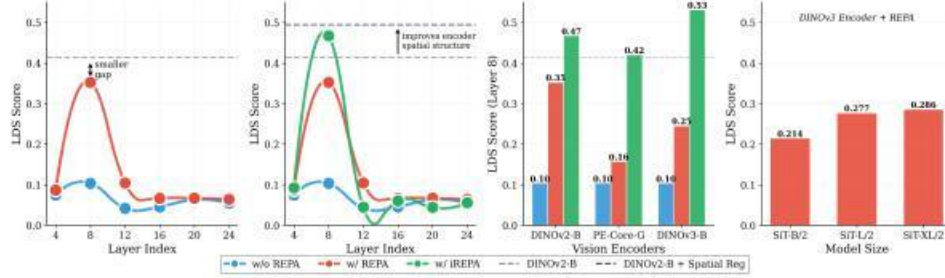


Figure 17: Representation alignment as bridging the spatial feature gap. a) We find that representation alignment can also be seen as bridging the spatial feature gap between diffusion and vision encoder patch features. b) iREPA (§4) accentuates the spatial features of the vision encoder (at cost of some global information) — helping achieve better generation performance. c) iREPA helps consistently improve performance across different vision encoders. d) Spatial structure improvements scale with model size. All results are reported with SiT-XL/2 at 400k iterations.

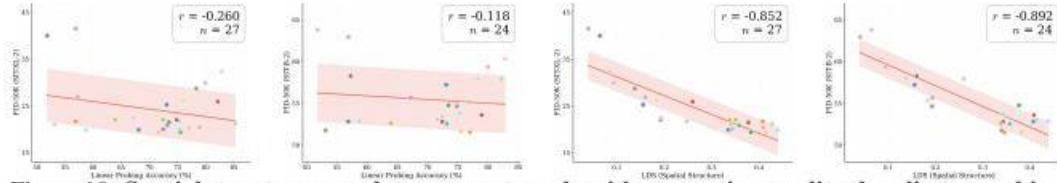


Figure 18: Spatial structure correlates more strongly with generation quality than linear probing. Correlation analysis across 27 vision encoders. **Left two:** Linear probing accuracy vs FID for SiT-XL-2 (Pearson $r = -0.26$) and SiT-B-2 ($r = -0.12$) shows weak correlation across model scales. **Right two:** Spatial structure (LDS) vs FID for SiT-XL-2 ($r = -0.85$) and SiT-B-2 ($r = -0.89$) shows strong correlation in the same settings. See Figure 10 in Appendix for detailed plots with encoder labels.

K ADDITIONAL ANECDOTAL COMPARISONS

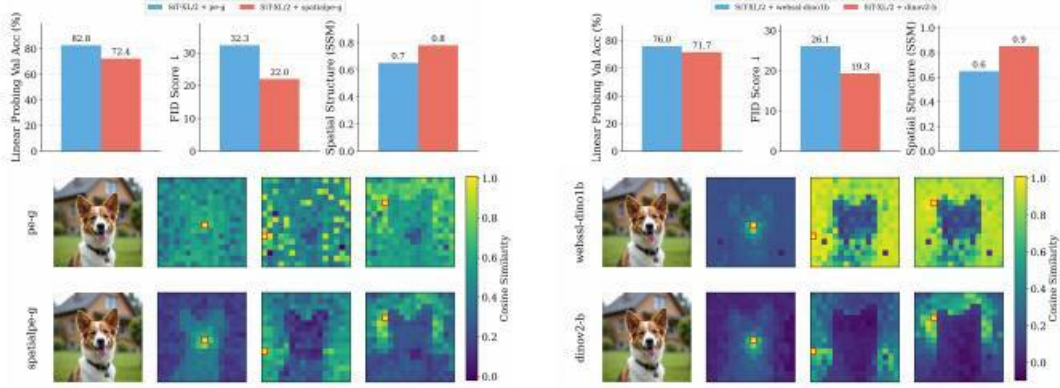


Figure 19: **Motivating anecdotes from recent SSL encoders — spatial structure matters.** **Top row:** Metrics comparison showing inverse relationship between ImageNet accuracy and generation quality. **Left:** PE-g achieves higher accuracy (82.8% vs 72.4%) but worse FID (32.3 vs 22.0) than Spatial-PE-g, with much lower spatial structure (LDS: 0.1 vs 0.4). **Right:** WebSSL-dino1b shows higher accuracy (76.0% vs 71.7%) but worse FID (26.1 vs 19.3) than DINOv2-b, with weaker spatial structure (LDS: 0.2 vs 0.4). **Bottom row:** Spatial cosine similarity visualizations confirm that encoders with better generation (Spatial-PE-g, DINOv2-b) maintain clear spatial coherence, while those optimized for classification (PE-g, WebSSL) lose spatial structure.

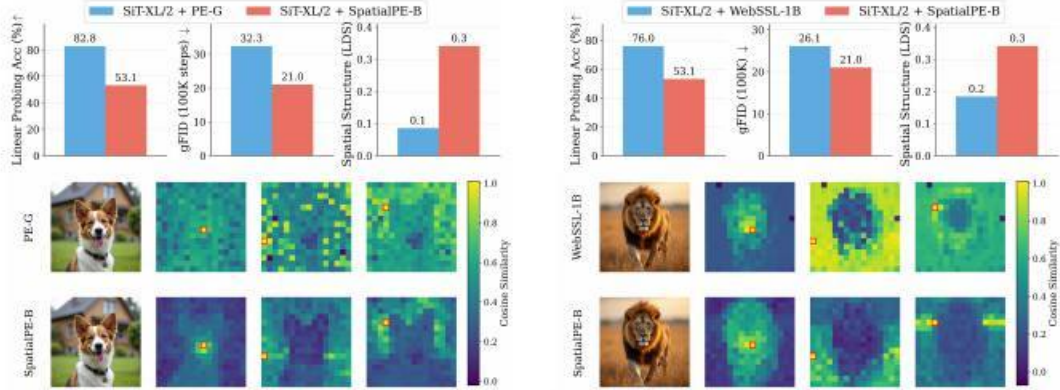


Figure 20: **Motivating examples — spatial structure matters.** **Top:** Metrics comparison showing inverse relationship between ImageNet accuracy and generation quality. **Left:** PE_{core}-G, despite having significantly higher validation accuracy (82.8% vs. 53.1%), shows worse generation quality compared to PE_{Spatial}-B (Bolya et al., 2025). **Right:** Similarly, WebSSL-1B (Fan et al., 2025) also shows much better global performance (76.0% vs. 53.1%), but worse generation. **Bottom:** We find that spatial structure instead provides a better predictor of generation quality than global performance. See §3 for spatial structure metric. All results reported at 100K using SiT-XL/2 and REPA.

L FURTHER DISCUSSION

Role of Spatial Normalization We visualize how spatial normalization changes vision encoder representations. Across examples, vision encoders can contain global components or overlays that reduce contrast between spatial tokens. For instance, tokens from one semantic region can be highly correlated with tokens from unrelated regions (e.g., background), making the features less spatially distinctive. Spatial normalization reduces this global overlay and increases contrast among spatial tokens, helping preserve local spatial organization while limiting global information that can interfere with generation quality.

The following figures show this effect across multiple examples, illustrating how spatial normalization shifts the representations toward clearer spatial structure:

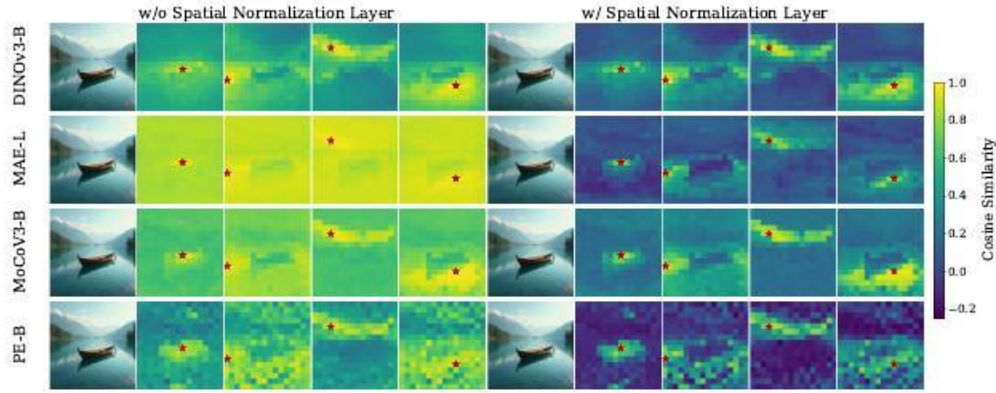


Figure 21: Visualizing the impact of spatial normalization (Example 1). The heatmaps compare token similarity before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure becomes clearer as global interference is reduced, yielding more distinct semantic boundaries.

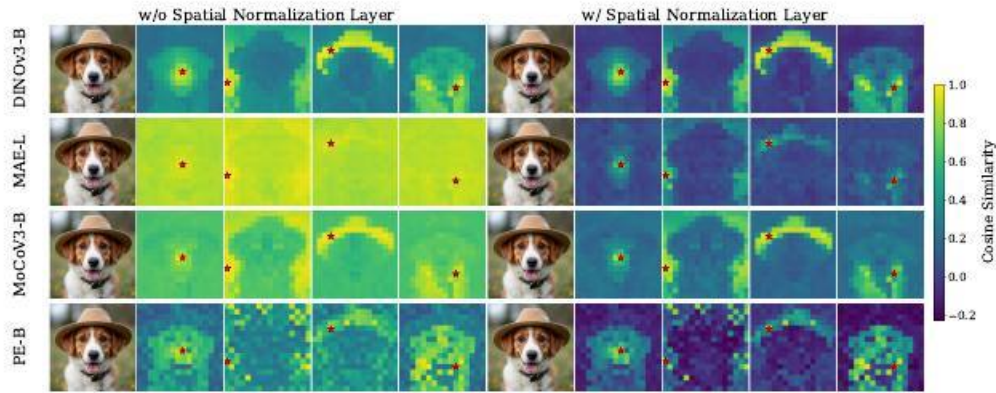


Figure 22: Visualizing the impact of spatial normalization (Example 3). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure becomes clearer as global interference is reduced, producing more distinct semantic boundaries.

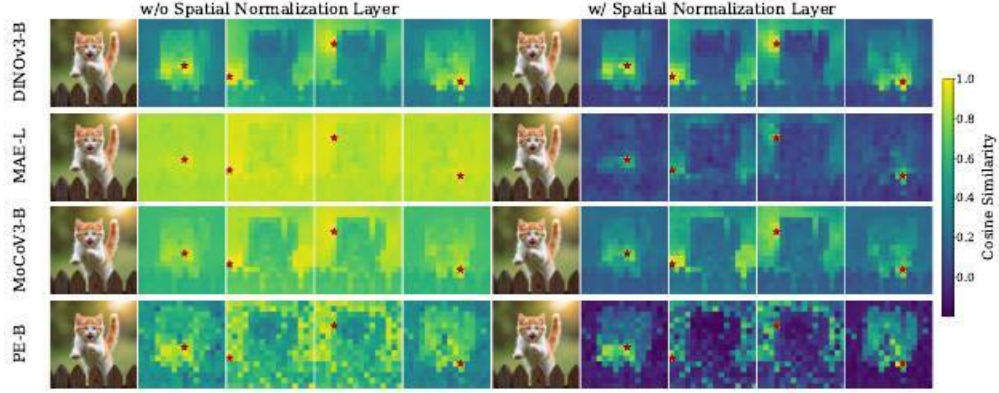


Figure 23: Visualizing the impact of spatial normalization (Example 2). Feature similarity maps show how spatial normalization improves spatial contrast. The original features (left) contain a global overlay that weakens the distinction between foreground and background regions. After normalization (right), spatial tokens are more locally coherent and separate semantic regions more clearly.

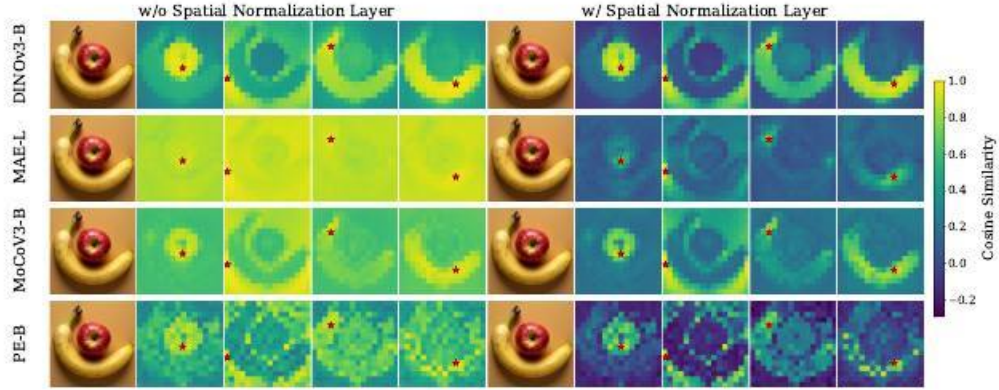


Figure 24: **Visualizing the impact of spatial normalization** (Example 4). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure becomes clearer as global interference is reduced, producing more distinct semantic boundaries.

M ADDITIONAL RESULTS

M.1 ADDITIONAL RESULTS AT HIGHER RESOLUTIONS

Setup. We conduct additional experiments on ImageNet-512 (Deng et al., 2009) to evaluate whether the proposed spatial improvements transfer to a higher-resolution generation setting. We follow the REPA (Yu et al., 2024) setup and report results across pretrained encoders (DINOv2, DINOv3, WebSSL, PE, etc.). All results use SiT-XL and SiT-B with 50 NFE at inference and no classifier-free guidance. Fig. 25 compares REPA before and after applying the spatial improvements (iREPA). The results show that the spatial improvements also transfer to higher resolution. Consistent gains are observed across the evaluated pretrained encoders (DINOv2, DINOv3, WebSSL, PE, etc.).

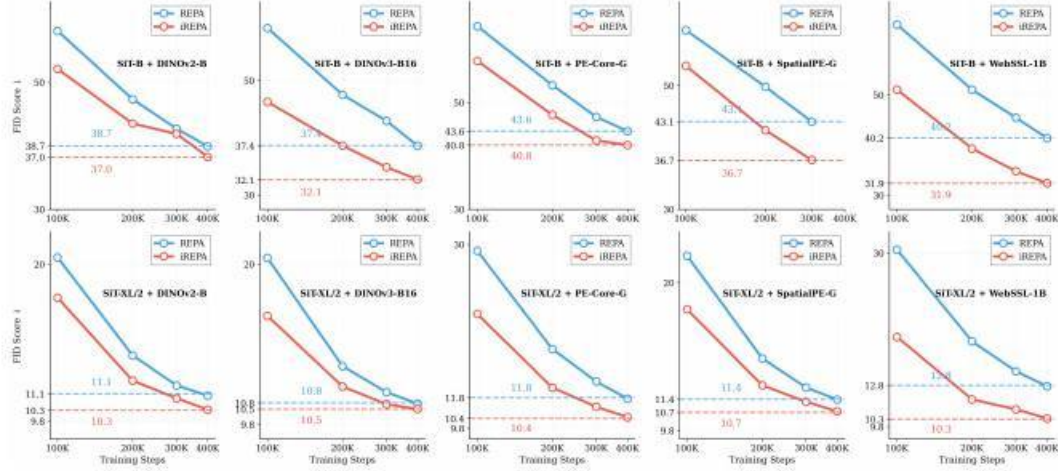


Figure 25: **Convergence results at 512x512 resolution.** Accentuating spatial features improves convergence across the evaluated resolutions, including Imagenet 256 (Figure 7) and Imagenet 512 (above).

M.2 ADDITIONAL RESULTS ON MULTIMODAL T2I TASKS

Setup. To test whether the spatial improvements extend beyond ImageNet class-conditional generation, we also evaluate multimodal T2I tasks. Following REPA (Yu et al., 2024), we use MMDiT (Esser et al., 2024) as the diffusion backbone and apply REPA with pretrained vision encoders (DINOv2, CLIP, WebSSL, PE, etc.). As in REPA (Yu et al., 2024), all models are trained for 150K steps with a batch size of 256 and evaluated using an ODE sampler with 50 NFE. Fig. 26 compares REPA before and after applying the spatial improvements (iREPA). The spatial improvements also transfer to multimodal T2I tasks, with consistent gains across the evaluated pretrained encoders (DINOv2, CLIP, WebSSL, PE, etc.).

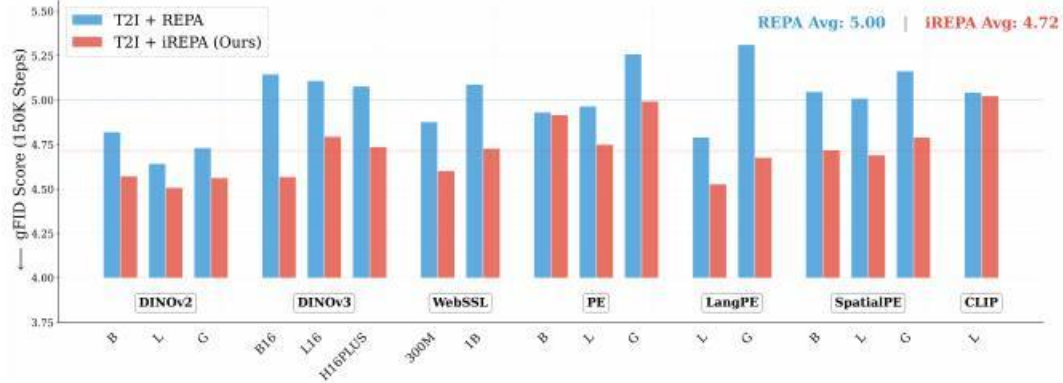


Figure 26: **Text-to-image generation across encoder variants.** Accentuating transfer of spatial features from the target representation to diffusion features improves convergence speed for both ImageNet (see §4) and multimodal T2I tasks (above). Consistent gains are observed across the evaluated pretrained encoders (DINOv2, CLIP, WebSSL, PE, etc.).

M.3 CORRELATION ANALYSIS WITHOUT OUTLIERS (MoCOv3-L AND MAE-L)

We repeat the correlation analysis from Section 3 after removing MoCOv3-L and MAE-L, which were identified as outliers. Figure 28 reports the updated correlations across model sizes. Spatial structure remains much more correlated with generation performance than linear probing accuracy. After removing the outliers, linear probing shows a small positive correlation with gFID (i.e., as linear probing performance increases, generation becomes worse). This trend is consistent with Sec. 2: target representations with higher global semantic performance can still show similar or worse generation performance under REPA.

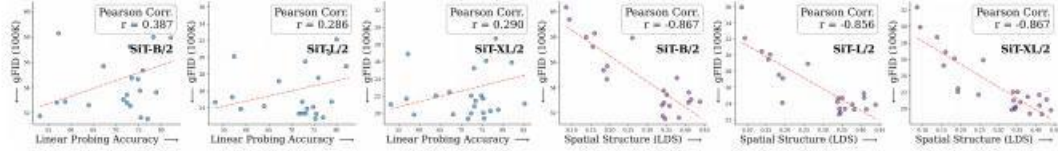


Figure 27: Correlation analysis without outliers. Across all model sizes (B, L, XL), spatial structure remains much more correlated with generation performance (Pearson $|r| > 0.85$) than linear probing accuracy (Pearson $|r| < 0.38$) after removing the outliers (MoCOv3-L and MAE-L). After removing the outliers, linear probing shows a small positive correlation with gFID (i.e., as linear probing performance increases, generation becomes worse). This trend is consistent with Sec. 2: target representations with higher global semantic performance can still show similar or worse generation performance under REPA.

M.4 ADDITIONAL QUALITATIVE RESULTS



Figure 28: **Additional Qualitative Results** comparing generation outputs before and after applying the spatial improvements to REPA. All results use PE-G (Bolya et al., 2025) as the pretrained vision encoder, 400K steps (80 epochs), and classifier-free guidance scale 4.0. Consistent with the quantitative results (§4), the spatial improvements (iREPA) improve the visual quality and coherence of the generated outputs.

M.5 ADDITIONAL RESULTS WITH ALTERNATIVE EVALUATION METRICS

Beyond the traditional evaluation metrics (Inception Score, FID, sFID, Precision, Recall), we test whether the findings are robust under an alternative generation-quality metric (Stein et al., 2023; Yang et al., 2023; Kynkäänniemi et al., 2023; Jayasumana et al., 2024). Specifically, we use CMMD (Jayasumana et al., 2024) with the PyTorch implementation from (Paul, 2024); the standard reference set from the OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021) is used as reference images. We then evaluate whether the key findings from §3 and §4 hold under CMMD.

Spatial structure correlates more strongly with generation performance than linear probing.

To test the robustness of §3, we repeat the large-scale correlation analysis across vision encoders using CMMD (Jayasumana et al., 2024). Results are shown in Figure 29. All results use SiT-B/2 (100K steps) and REPA. All spatial metrics correlate more strongly (Pearson $|r| > 0.88$) with generation performance under CMMD than linear probing does (Pearson $|r| = 0.074$), showing that the empirical findings from §3 are robust to the choice of FID or CMMD.

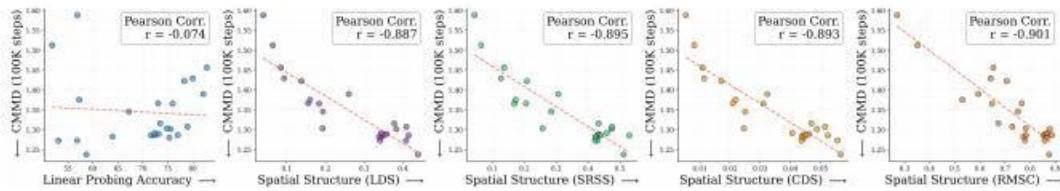


Figure 29: **Correlation analysis with CMMD metric.** We repeat the correlation analysis from §3 using CMMD (Jayasumana et al., 2024) instead of gFID. All results use SiT-B/2 (100K steps) and REPA. Spatial metrics correlate more strongly (Pearson $|r| > 0.88$) with CMMD generation performance than linear probing does (Pearson $|r| = 0.074$). This shows that the key empirical findings from §3 are robust to the evaluation metric.

Accentuating transfer of spatial features helps improve generation quality. We next test whether the §4 conclusions hold under CMMD (Jayasumana et al., 2024) as well as traditional metrics (gFID, sFID, IS, etc.). Table 8 shows that, across vision encoders, accentuating spatial transfer with iREPA improves convergence under both CMMD and traditional evaluation metrics (IS, FID, sFID, Prec., Recall). The improvements appear both with and without classifier-free guidance (CFG).

Vision Encoder	Steps	w/o CFG						w/ CFG					
		CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑	CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑
DINOv2-B	100K	0.702	69.20	19.3	5.89	0.64	0.61	0.529	157.2	6.35	5.91	0.77	0.54
+iREPA	100K	0.652	77.92	16.9	6.26	0.66	0.61	0.484	179.3	5.15	6.23	0.78	0.54
DINOv2-B	400K	0.455	127.4	7.76	5.06	0.70	0.66	0.320	263.0	1.98	4.60	0.80	0.61
+iREPA	400K	0.438	128.6	7.52	4.89	0.71	0.65	0.310	268.8	1.93	4.59	0.80	0.60
DINOv3-B	100K	0.749	63.64	21.4	6.14	0.63	0.60	0.571	144.0	7.57	6.09	0.76	0.53
+iREPA	100K	0.651	78.79	16.2	6.14	0.66	0.61	0.481	181.9	4.87	6.10	0.78	0.55
DINOv3-B	400K	0.474	126.7	8.10	5.06	0.70	0.66	0.336	261.2	1.99	4.58	0.80	0.61
+iREPA	400K	0.441	132.9	7.13	4.93	0.71	0.66	0.314	272.4	1.89	4.58	0.80	0.60
WebSSL-1B	100K	0.825	53.87	25.5	6.57	0.61	0.59	0.622	124.0	9.59	6.37	0.76	0.52
+iREPA	100K	0.653	77.47	16.6	6.18	0.66	0.61	0.484	177.6	5.09	6.11	0.79	0.54
WebSSL-1B	400K	0.512	116.9	9.39	5.14	0.70	0.64	0.354	250.5	2.24	4.61	0.81	0.58
+iREPA	400K	0.445	130.8	7.48	4.91	0.70	0.65	0.311	271.6	1.90	4.58	0.80	0.61
PE-Core-G	100K	0.922	42.74	32.3	6.70	0.57	0.59	0.714	97.2	14.1	6.56	0.71	0.53
+iREPA	100K	0.697	75.01	18.1	6.03	0.64	0.61	0.525	176.8	5.66	6.08	0.77	0.54
PE-Core-G	400K	0.525	109.4	10.4	5.00	0.69	0.64	0.366	238.4	2.44	4.57	0.81	0.59
+iREPA	400K	0.458	132.0	7.78	5.02	0.70	0.65	0.322	275.4	1.93	4.59	0.80	0.61
CLIP-L	100K	0.790	54.07	25.2	6.65	0.61	0.60	0.605	124.5	9.77	6.59	0.75	0.52
+iREPA	100K	0.657	74.46	17.8	6.33	0.65	0.61	0.489	172.0	5.67	6.33	0.78	0.54
CLIP-L	400K	0.487	117.8	8.97	4.98	0.70	0.65	0.339	258.7	2.15	4.64	0.81	0.59
+iREPA	400K	0.442	130.1	7.42	5.03	0.71	0.65	0.307	271.8	1.96	4.64	0.80	0.61

Table 8: **Additional results with alternative evaluation metrics.** We provide additional results with the CMMD metric (Jayasumana et al., 2024). All results are reported using SiT-XL/2, 256 batch size and traditional REPA as baseline. We adopt the pytorch implementation from (Paul, 2024) for computing the CMMD metric. Standard reference set from OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021) is used as reference images. Across various vision encoders, accentuating transfer of spatial features (iREPA) helps improve convergence speed with both CMMD and traditional evaluation metrics (IS, FID etc.). This demonstrates that key empirical findings from §4 are robust to the choice of evaluation metric.

M.6 SPATIAL IMPROVEMENTS WITH SAM2

Table 9 and Figure 30 examine spatial normalization (§4) on SAM2-S (46M) encoder features². As shown in Table 9, for SAM2, spatial normalization improves performance, but the gains are marginal. This is expected because spatial normalization improves contrast by removing the global component (mean patch token), whereas SAM2 already contains little global information (validation accuracy < 24%). As a result, spatial normalization only slightly increases spatial contrast (Figure 30).

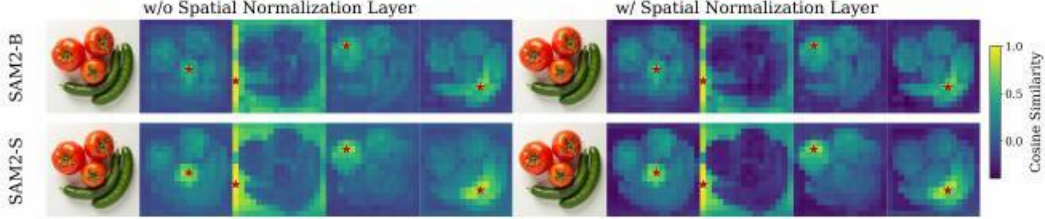


Figure 30: **Impact of spatial normalization on SAM2.** We observe that for SAM2, while use of spatial normalization layer does help enhance the spatial contrast, the improvements can be marginal. This is because spatial normalization (§4) relies on removing the global component (mean of patch tokens) to enhance spatial contrast. Since SAM2 already has little to no global information (validation accuracy < 24%), spatial normalization only slightly improves the spatial contrast.

Vision Encoder	Steps	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
SAM2-S	100K	50.69	25.32	6.52	0.631	0.588
+iREPA	100K	53.14	24.52	6.28	0.629	0.591
SAM2-S	400K	110.77	9.54	4.93	0.699	0.638
+iREPA	400K	114.62	9.10	4.89	0.704	0.640

Table 9: **Impact of spatial improvements on SAM2.** All results use SiT-XL/2 without classifier-free guidance, SAM2-S (46M) as the vision encoder, and standard REPA as the baseline.

²Note that similar to (Bolya et al., 2025), we use the intermediate output of vision encoder for SAM2 features and not the mask logits. As shown in (Bolya et al., 2025), while mask logits lead to sharper spatial maps, the mask logits itself are not suitable as a target representation.

M.7 ADDITIONAL RESULTS WITH FULL-FINETUNING ACCURACY INSTEAD OF LINEAR PROBING ACCURACY

Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. We repeat the §3 correlation analysis using validation accuracy after full-finetuning³ instead of linear probing. Results are shown in Fig. 31. All results use SiT-XL/2 (100K steps) and REPA. To obtain full-finetuning validation accuracy, we fine-tune the vision encoder with learning rate $5e-5$, warmup ratio 0.1, and 30 total epochs. Spatial metrics correlate more strongly with generation performance than full-finetuning accuracy. gFID also shows a weak positive correlation with validation accuracy after full-finetuning (Pearson $r = 0.317$), meaning that as validation accuracy increases, gFID increases and generation performance worsens. This is consistent with §2: target representations with higher global semantic performance can still show similar or worse generation performance with REPA.

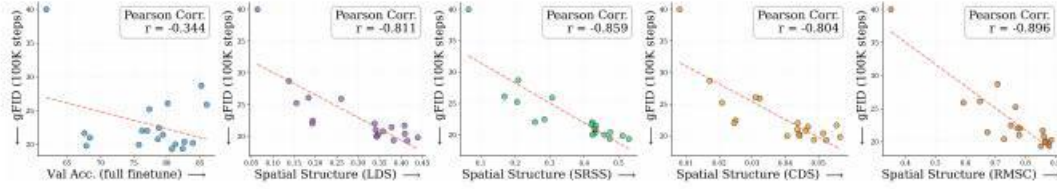


Figure 31: Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. We repeat the correlation analysis from §3 using validation accuracy after full-finetuning instead of linear probing. All results use SiT-XL/2 (100K steps) and REPA. Spatial metrics correlate more strongly with generation performance than full-finetuning accuracy. gFID shows a weak positive correlation with validation accuracy after full-finetuning (Pearson $r = 0.317$), meaning that as validation accuracy increases, gFID increases and generation performance worsens. This is consistent with §2: target representations with higher global semantic performance can still show similar or worse generation performance with REPA.

³Please note that while the final findings remain similar, in context of REPA linear probing might be more accurate for estimating global information in “pretrained” vision encoders. This is because REPA uses the “pretrained” encoder features themselves for regularization. Finetuning the encoder itself can impact the amount of global information. Same as REPA (Yu et al., 2024), we therefore use linear probing accuracy as default for measuring global information.

M.8 ADDITIONAL RESULTS ON PIXEL-SPACE DIFFUSION MODELS

Method	#Params	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
PixelFlow (Chen et al., 2025)	459M	24.67	54.33	9.71	-	0.58
PixDDT (Wang et al., 2025b)	434M	36.24	46.37	17.14	-	0.63
PixNerd (Wang et al., 2025a)	458M	43.01	37.49	10.65	-	0.62
DeCo w/o $\mathcal{L}_{\text{FreqFM}}$	426M	46.44	34.12	10.41	-	0.64
DeCo + REPA (Ma et al., 2025b)	426M	48.35	31.35	9.34	-	0.65
JiT (Li & He, 2025a)	459M	29.37	49.06	11.21	0.40	0.62
JiT+iREPA (Ours)	459M	50.72	29.19	9.42	0.51	0.65

Table 10: **Generation performance of pixel-space diffusion models.** Despite its simplicity, when combined with JiT (Li & He, 2025a) iREPA outperforms recently proposed state-of-art pixel-space diffusion methods e.g., DeCo (Ma et al., 2025b). All results are reported with 200K training iterations with a batch size of 256 and evaluated using 50-step Euler sampling without classifier-free guidance. Results for DeCo are obtained directly from (Ma et al., 2025b).